

Deepak Mallampati

+1-330-554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | github.com/Deepak0070 |
Kent, OH, USA - F-1 OPT, open to sponsorship; remote-first

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) shipping agentic LLM workflows, multi-agent pipelines, and RAG systems end-to-end - from data preprocessing through agent design, evaluation, and FastAPI/Docker deployment. Track record: ~35% retrieval precision gain, >30% hallucination reduction, ~30% post-deploy defect reduction. Comfortable owning the deployment journey: first process file to a working improving production workflow, debugging integration failures, writing custom agents where platform defaults don't fit, and feeding patterns back to product. Targeting Forward Deployed and Applied AI Engineer roles (Camunda ProcessOS and similar).

SKILLS

Agentic AI Workflows • Multi-Agent Systems • RAG + Grounding • LLM Application Dev • Python / FastAPI • Docker / Containers • Legacy System Integration • Structured Outputs + Evals • Customer-Facing Delivery • Production Observability

Languages: Python, TypeScript, JavaScript, SQL (T-SQL, PostgreSQL, SQLite), Java (ramping), C++

AI / ML: PyTorch, scikit-learn, XGBoost, LangChain, LlamaIndex, OpenAI, Anthropic, Hugging Face Transformers, Diffusers, YOLOv8

LLM / GenAI: RAG, agentic workflows, multi-agent orchestration, schema-validated JSON contracts, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search

Backend: FastAPI, Flask, REST, WebSockets, Docker, Celery, Playwright

Frontend: React, Next.js, Vite, Tailwind, Chrome Extensions (Manifest V3)

Data: PostgreSQL + pgvector, MongoDB, Redis, Pandas, NumPy, Snowflake

DevOps: Docker, Jenkins, GitHub Actions, Grafana, Prometheus, AWS (EC2, S3, Lambda); legacy-API integration patterns; structured release validation + rollback checkpoints

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Agentic Orchestration, Workflow Automation, Regulated Industries

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Agent Engineering, Production LLM

USA - Healthcare Technology, AI Consulting

- Built **Retrieval-Augmented Generation (RAG)** for clinical knowledge retrieval; recursive semantic chunking and transformer embeddings improved retrieval precision ~**35%**, reduced irrelevant retrieval >**30%**, response alignment >**90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care navigation, documentation) with structured reasoning, tool discipline, and grounding rules; agent response stability ~**25%**, hallucinations cut >**30%**.
- Designed a 5-stage **multi-agent Healthcare Claims pipeline** (Intake → Validation → Consistency → Duplicate → Fraud Risk) with **schema-validated JSON contracts** between agents to prevent cascading hallucinations; audit-ready traces for clinical reviewers.
- Packaged ML/LLM inference as **FastAPI / Flask** services on **Docker** with structured logging and load simulation; post-deploy defects ~**30%**.
- EHR preprocessing (Pandas, NumPy); dataset reliability >**98%**, downstream model instability **-40%**.
- HIPAA-conscious governance: de-identification, data lineage, evaluation audit trails, stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Enterprise ERP Integration

Hyderabad, India - Oil & Gas ERP, regulated

- Optimized **T-SQL stored procedures** on the Synthesis Order-to-Cash ERP (contracts, nominations, allocations, invoicing); query time **-20%**, retrieval latency **-25%**.
- Built **Jenkins CI/CD pipelines** for schema updates and stored-procedure deployments with structured release validation and rollback checkpoints; deploy errors **-30%**, release cycle **-35 to -40%**.
- Designed performance dashboards with **SQL Server DMVs + Grafana** to track long-running queries, deadlocks, CPU/I/O contention; incident recurrence **-25%**, root-cause resolution **-30%**.
- Tuned index maintenance and partition-aware queries; reconciliation runtime **-15%**, deadlocks **-15%**.
- Supported **RBAC and audit logging** for financial modules in a compliance-sensitive environment - legacy-API integration discipline that travels to BPMN connector / custom-agent work.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer

Hyderabad, India - Nonprofit, Video Analytics, Applied NLP

- Built **transformer-based video summarization** over 5,000+ recorded sessions; hierarchical summarization and timestamp-aligned clip extraction cut manual review **60-70%**.
- AI-selected highlights aligned with human-curated moments **~85%**.
- Implemented **document Q&A with RAG**; hallucinations **~30%**, contextual accuracy **>85%**.
- Deployed as a **Flask API** with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

5-stage multi-agent pipeline (Intake → Validation → Consistency → Duplicate → Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policies. Audit-ready explainable risk scoring. Same architectural pattern as chaining agents reliably across BPMN steps.

Python, LangChain, FastAPI, pgvector, GPT-4, schema contracts

Manga Lens - Multi-Provider AI Vision Chrome Extension (shipped, Chrome Web Store)

Solo-built Chrome extension (Manifest V3, content scripts, service workers, captureVisibleTab) integrating 4 AI vision providers with per-provider payload normalization (WebP for cloud, JPEG for Ollama to avoid CUDA crash). 7-day translation cache, per-domain configs for 29 sites, narrowed host permissions, privacy policy. Evidence of shipping solo through ambiguous integration tradeoffs.

TypeScript, Manifest V3, Service Workers, OpenAI / Anthropic / Google / Ollama

Dream Decoder - Multimodal Creative Platform

End-to-end multimodal pipeline: text description, intermediate prompt transformation, grounded image generation. ~30% contextual alignment gain, ~25-30% first-pass image success.

FastAPI, React + Vite, Tailwind, Perplexity Sonar, Replicate

YOLOv8 Driver Drowsiness Detection

Real-time computer vision for driver drowsiness; sub-50ms inference on edge hardware. Walk-forward validation and class-weighted training for imbalanced safety data.

PyTorch, YOLOv8, OpenCV, ONNX

EDUCATION

Master of Science, Computer Science - Kent State University

Aug 2023 - May 2025

Focus: Applied Machine Learning, NLP, Database Systems.

Bachelor of Technology, Computer Science - KL University

Jun 2019 - May 2023

Founder / Lead of E-Farming platform (university project, pilot with 3 cooperatives).

CERTIFICATIONS

Deep Learning Specialization - DeepLearning.AI (Coursera)

2024